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Blockchain Assignment 1

Bitcoin Basics

1. Bitcoin was introduced by whom?
   Satoshi Nakamoto

2. Bitcoin transactions are recorded on which data structure?
   Blockchain

3. Bitcoin uses which consensus mechanism?
   Proof-of-Work (PoW)

4. The smallest unit of Bitcoin is called?
   Satoshi (1 BTC = 100,000,000 satoshis)

5. Bitcoin is a **peer-to-peer** type of what system?
   Peer-to-peer electronic cash system

Wallets

6. A wallet stores the user's Private key.

7. The recovery phrase used in wallets is called a Seed (or Mnemonic) phrase.

8. Hardware wallets store keys offline (online/offline).

9. Bitcoin addresses are derived from Public keys.

10. What is required to sign a transaction? (public/private) Private

Security

11. Bitcoin prevents double-spending using Blockchain (or Consensus).

12. Bitcoin is secured using Asymmetric cryptography (asymmetric/symmetric).

13. To run a Bitcoin node, the software required is Bitcoin Core.

14. The process of verifying transactions is called Validation.

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15. The Bitcoin network is described as decentralized.

Mining

16. Miners solve cryptographic puzzles to add blocks.
17. Mining rewards a block with newly created Bitcoins.
18. The block time target for Bitcoin is 10 minutes.
19. Mining difficulty adjusts every 2016 blocks.
20. Miners compete to find a valid block hash.

Blockchain Structure

21. Each block contains a Merkle root.
22. Blocks are linked using the previous block's hash.
23. A Bitcoin transaction contains inputs and outputs.
24. Unspent transaction outputs are called UTXOs.
25. The first-ever Bitcoin block is called the Genesis block.

Network

26. A full node stores the entire blockchain locally.
27. Bitcoin uses TCP protocol for communication.
28. A light wallet uses SPV mode.
29. Nodes relay information across a peer-to-peer network.
30. The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: 960831

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- Block hash:000000000000000000001f349c1d37e148d2e8f07852573c60abb0ed118992711
- Timestamp:2026-08-03 11:48:18 GMT +5.5
- Number of transactions:6118
- Miner/pool name:F2Pool

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction. 7387176e4f05396e7e36a7ee9b3af8525336b9a10de57ff83893bf4e36c6ea9a
11. Number of inputs:2
12. Number of outputs:2
13. Transaction fee:0.00001290 BTC (5.03 sat/vB)
14. How many confirmations does it have? 0 (Unconfirmed)

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.
16. Record:

- Total BTC received:27863.05123146 BTC
- Total BTC sent:27621.63683970 BTC
- Current balance:241.41439176 BTC
- What is the role of a full node?

  A full node stores the entire blockchain, validates transactions and blocks independently, and relays valid transactions and blocks to other nodes, helping maintain the security and decentralization of the Bitcoin network.

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List any two developer resources available on bitcoin.org.

- Bitcoin Developer Documentation
- Bitcoin Core Source Code (GitHub)

Explore : https://coinmarketcap.com/

Task C1: Market Information

- Current price of BTC:$63,501.57
- 24-hour % change:+2.71%
- Market capitalization:$1.27 trillion
- Circulating supply:20.06 million BTC
- Total/maximum supply:20.06 million BTC / 21 million BTC

Task C2: Historical Data

- What was the price of Bitcoin one year ago?
  Approximately **$58,970** per BTC.
- Note the highest and lowest prices in the last 30 days.

  Highest price (last 30 days): $73,562
  Lowest price (last 30 days): $58,562.44

Explore https://www.investopedia.com/

1.Why is Bitcoin considered volatile?

Bitcoin is considered volatile because its price can change rapidly due to factors such as supply and demand, investor sentiment, news, government regulations, institutional investment, and macroeconomic events. Since the cryptocurrency market operates 24/7, prices can fluctuate significantly at any time.

2.List two investment risks mentioned.

1. High price volatility.
2. Regulatory uncertainty

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- Part E: Reflection & Analysis

1.Which website gave you the most clear understanding? Why?

Blockstream Explorer gave me the clearest understanding because it presents detailed information about Bitcoin blocks, transactions, addresses, and the blockchain structure in an organized and easy-to-understand format. It helped me visualize how blocks, transactions, and confirmations are connected.

2.In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transactions in blocks that are linked together using cryptographic hashes. Every transaction is verified by network nodes through a consensus mechanism before being added to the blockchain. Once a block is added, altering its contents would change its hash and invalidate all subsequent blocks, making tampering extremely difficult. Since every participant maintains a copy of the blockchain and independently verifies transactions, no single organization controls the network, creating a transparent, secure, and decentralized system based on cryptography and consensus rather than trust in a central entity.